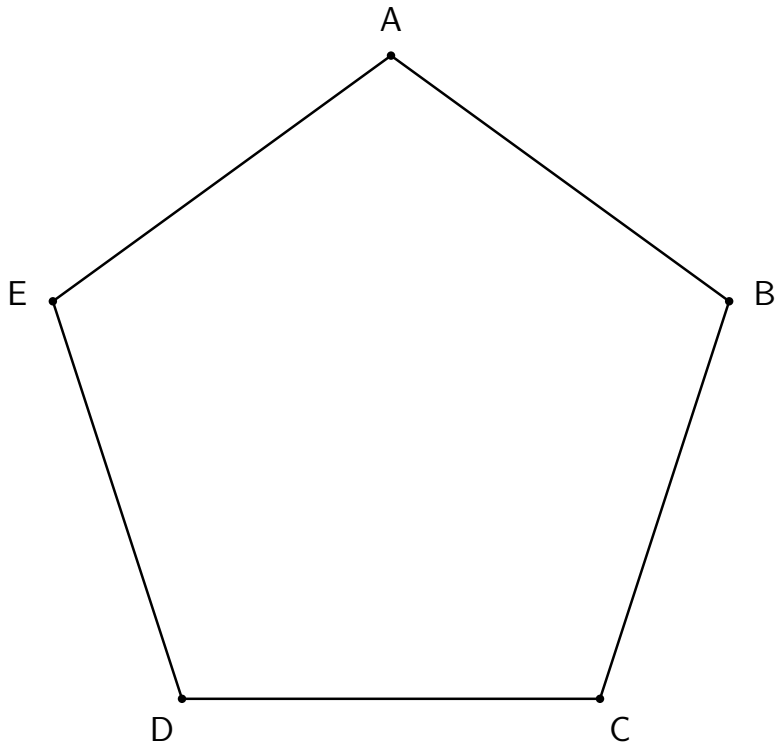
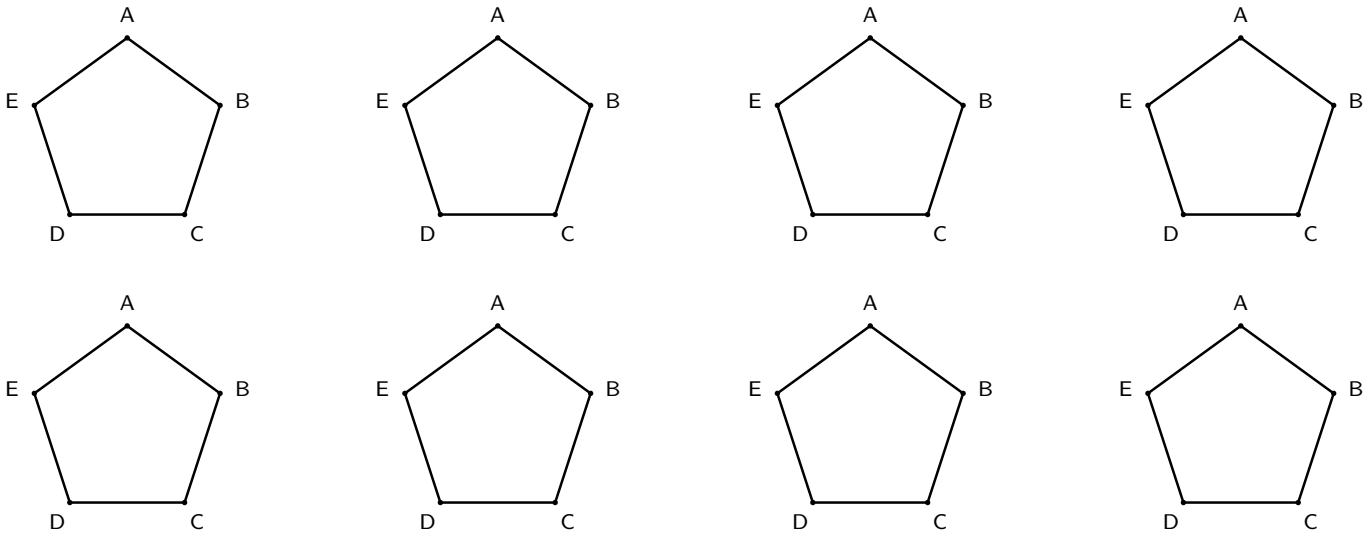
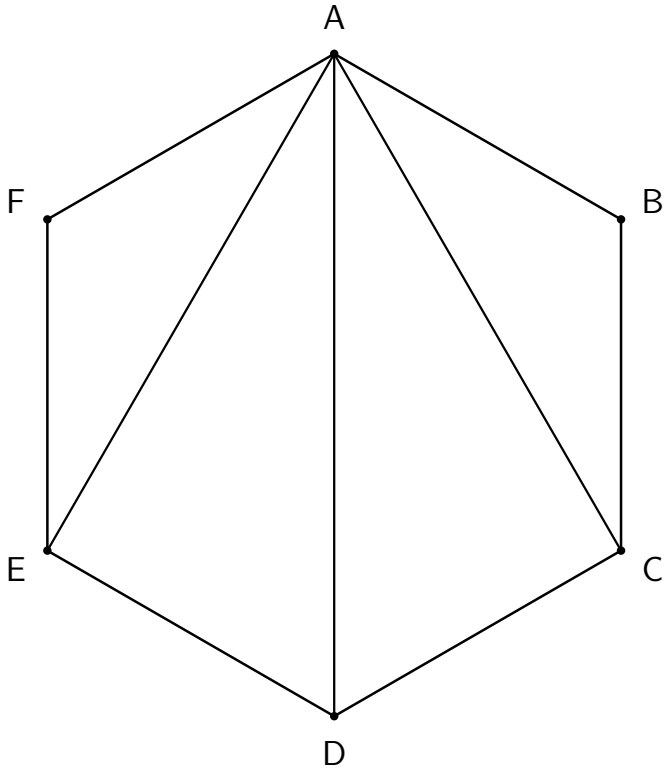
Fill the whole polygon with triangles, using straight lines between printed corners. Lines inside a polygon may meet at corners but may not cross. Keep the corner labels fixed, even if you turn the page. Use tracing paper to change printed lines.

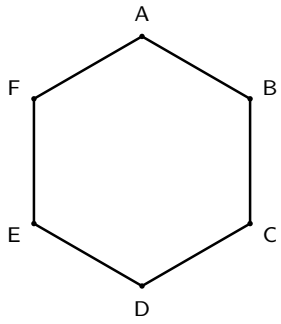
Problem 1: Find every filling of the five-corner polygon. Explain how your collection shows that none are missing.

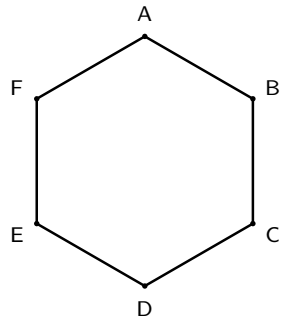


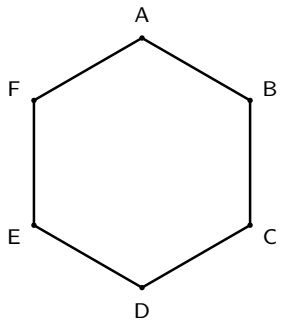


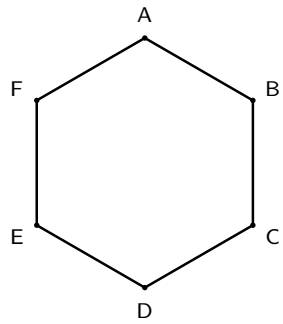
Problem 2: Erase one inside line and replace it with a different line so the polygon stays filled with triangles. Find every result of one such change from each drawing.

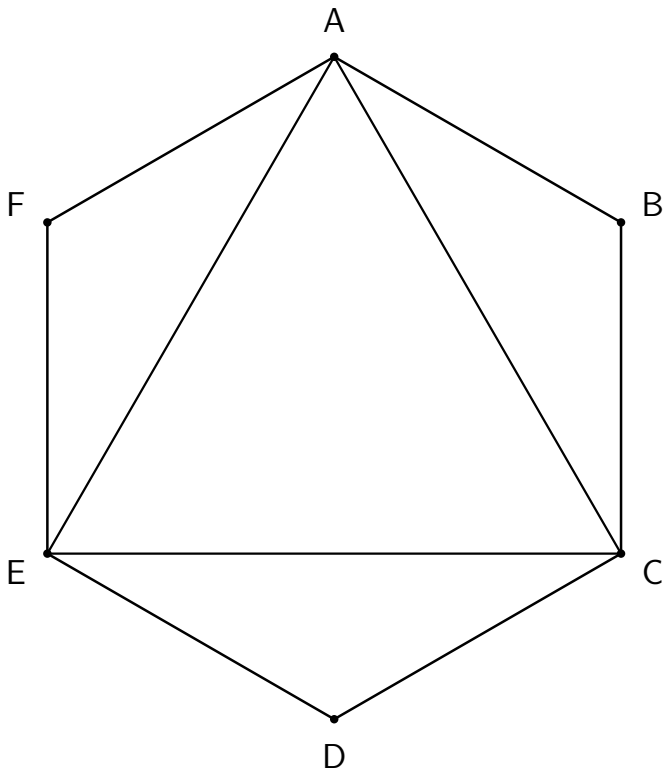


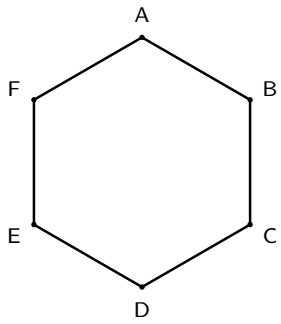


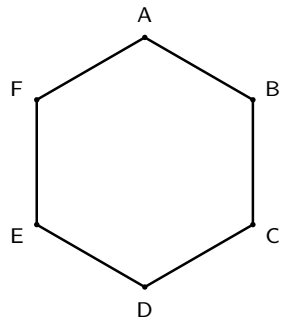


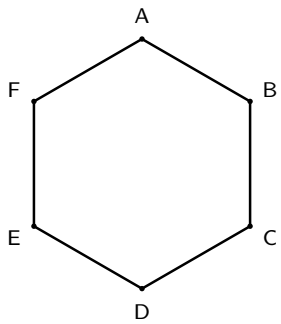


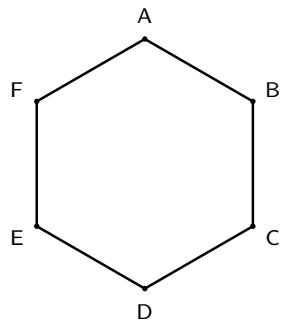




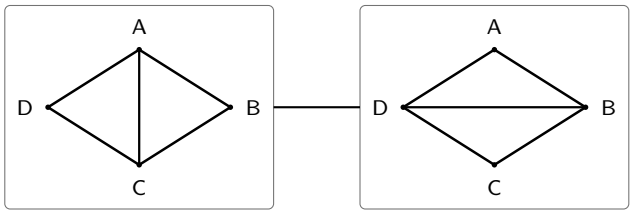




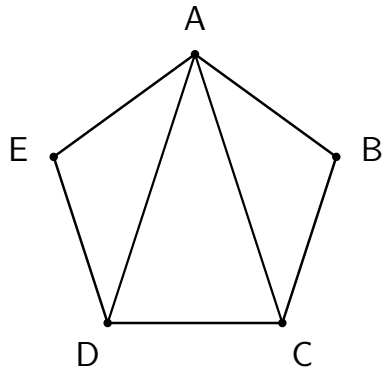


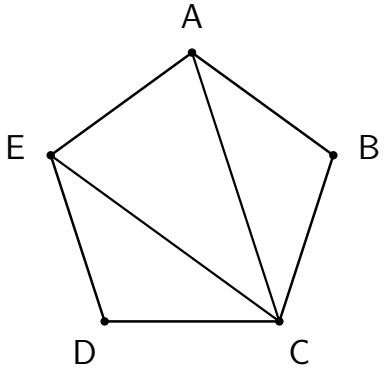


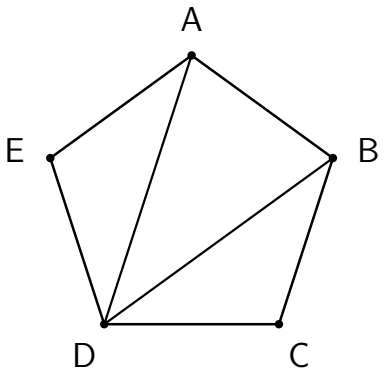
Problem 3: A change of one inside line that keeps the polygon filled with triangles is called a flip. Join two drawings when one flip changes one into the other. Include every possible join. Can you return to a filling after an odd number of flips? Show a route, or explain why it is impossible.

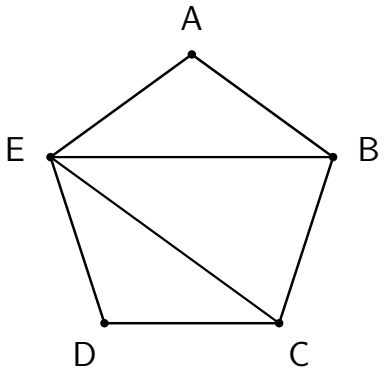


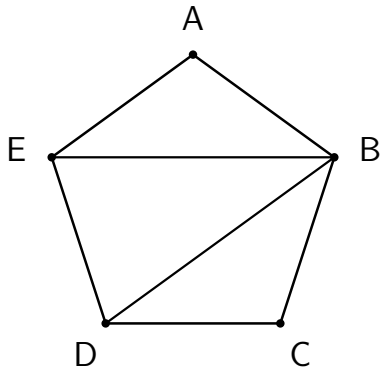
Example: erase AC, draw BD. The join connects whole fillings; it is not an inside line.



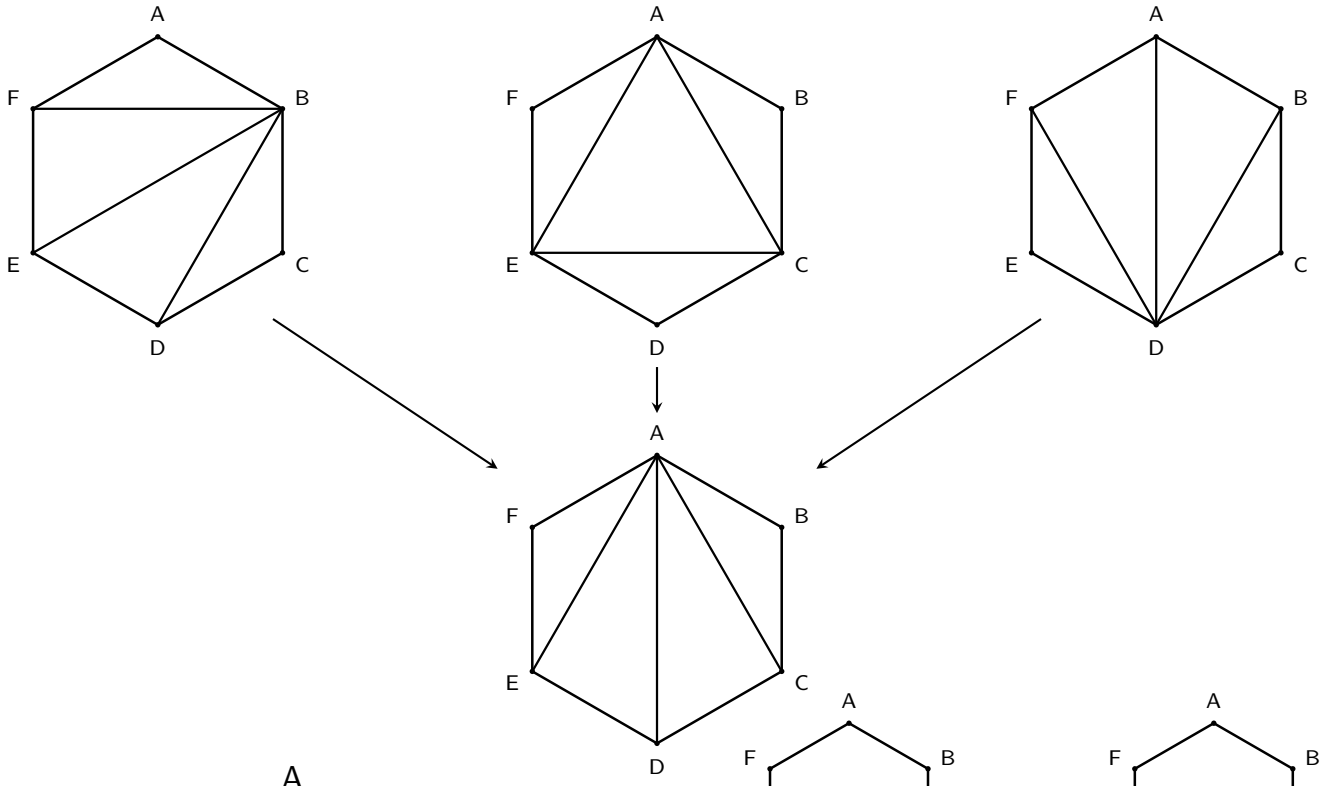


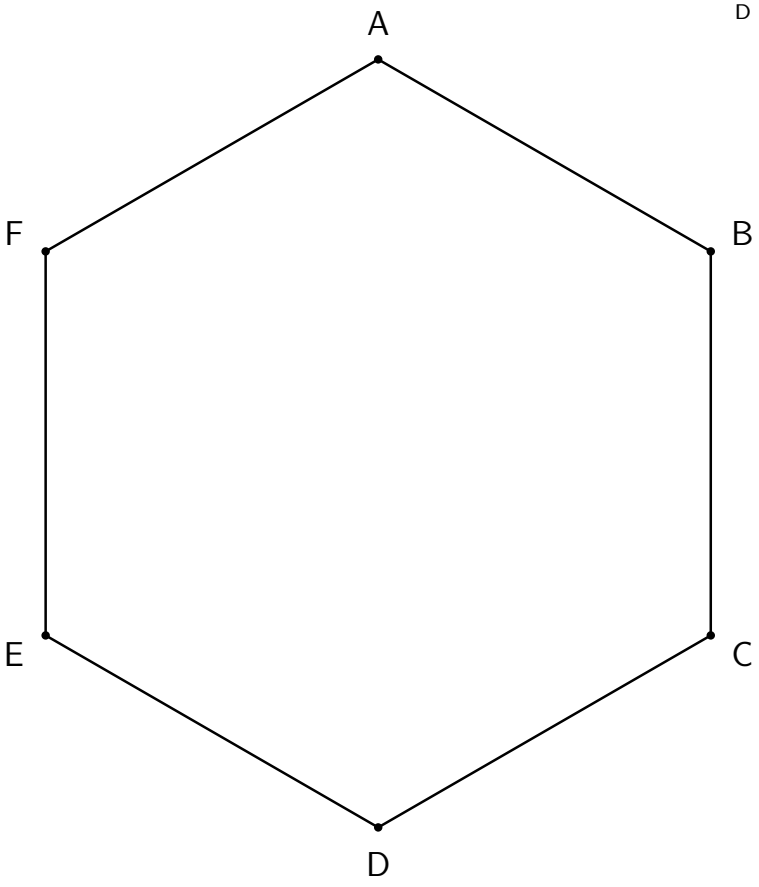


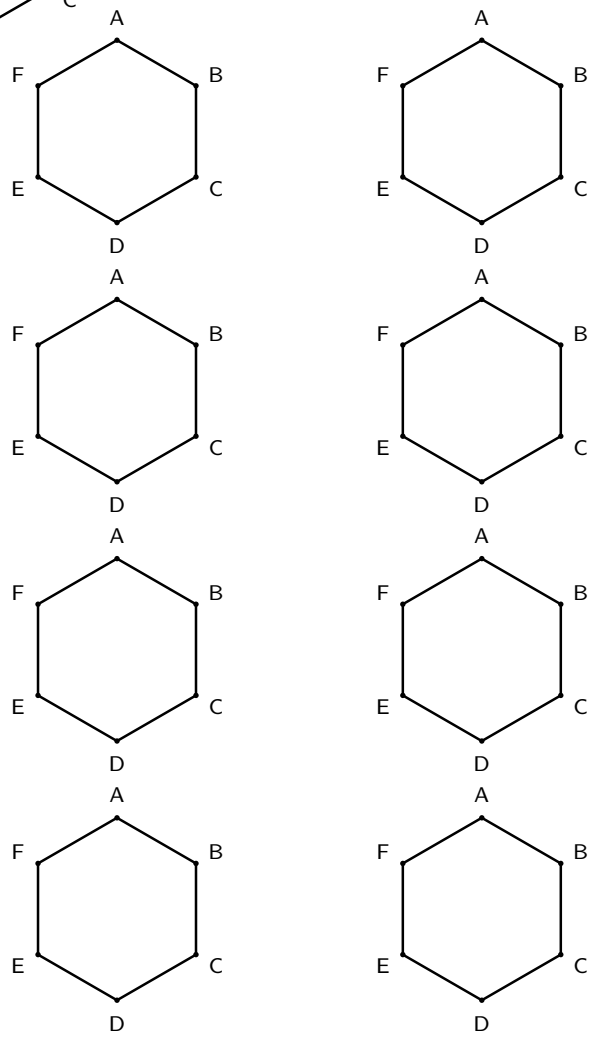




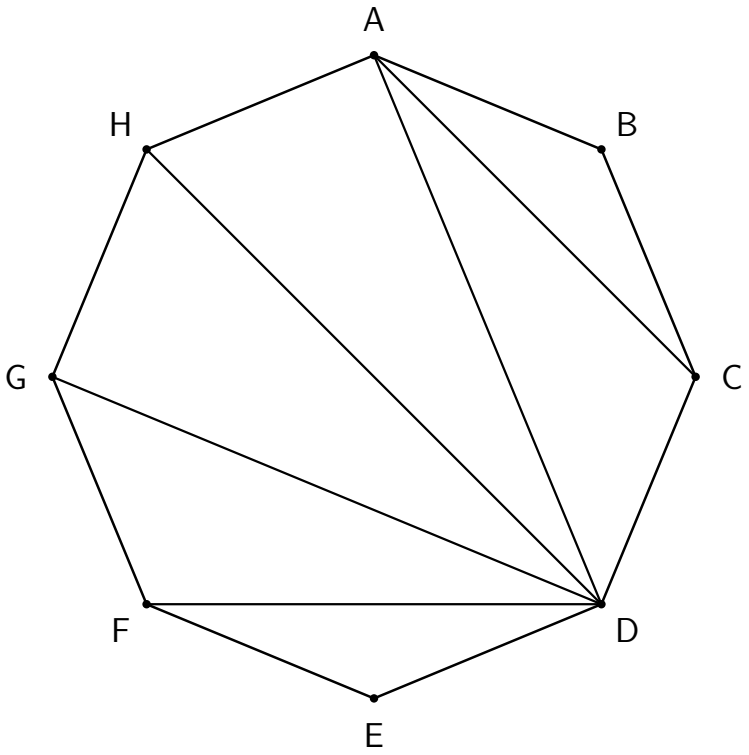
Problem 4: Change each of the three starting fillings into the fourth filling using as few flips as possible. Explain why no shorter routes can work.

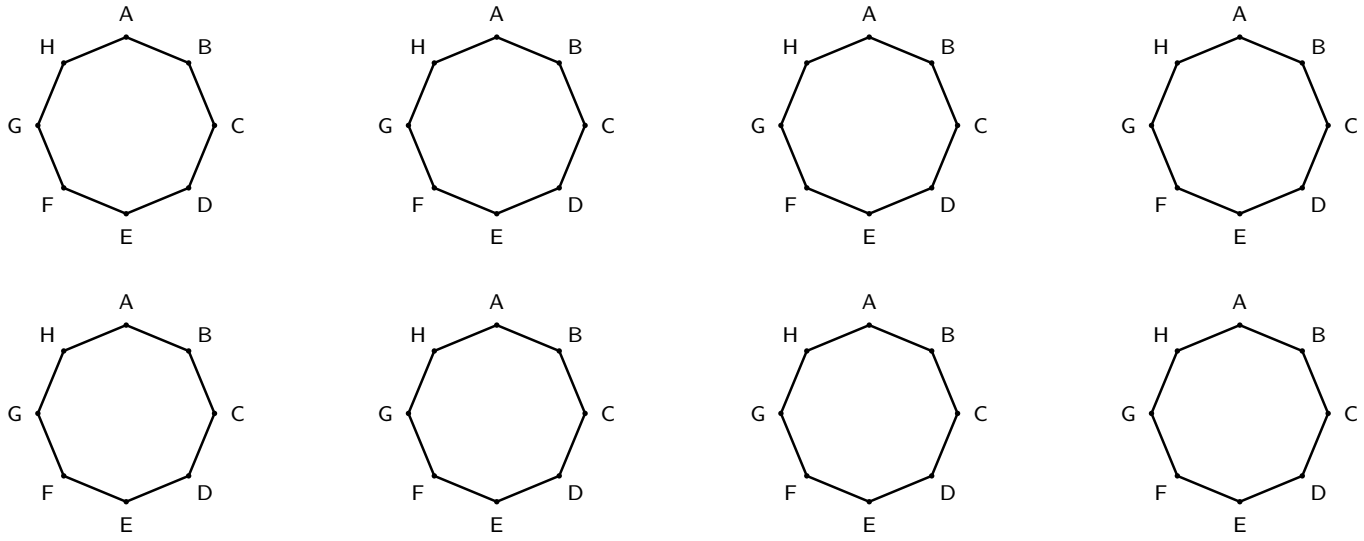




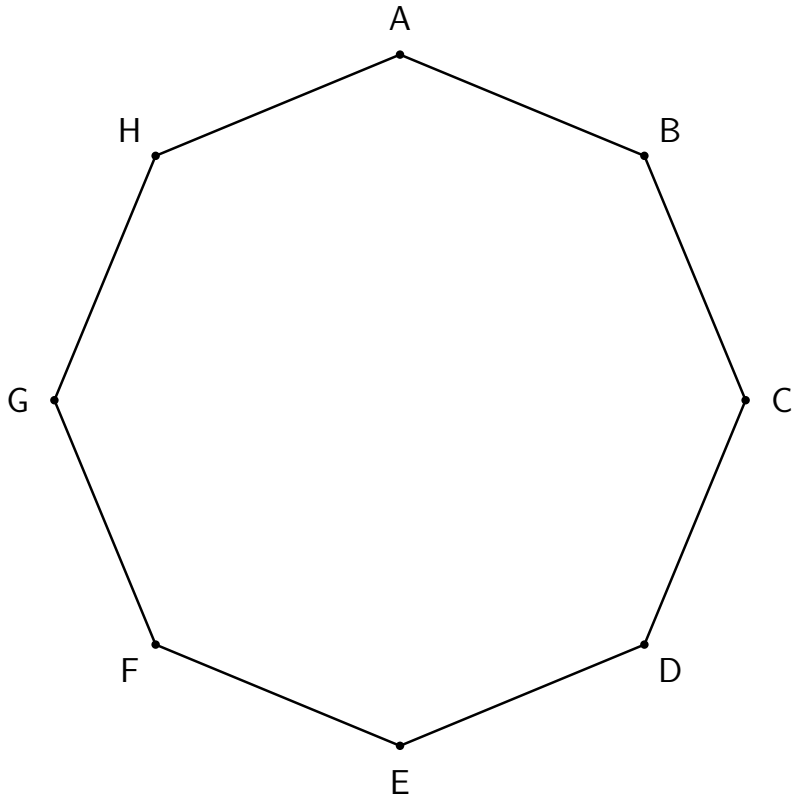


Problem 5: A filling whose inside lines all meet at one corner is called a fan. Find the fewest flips from this filling to a fan at A, and to a fan at E. Explain why each route is shortest.





Problem 6: Find a rule for the fewest flips needed to turn any filling into a fan at A. Explain why your rule works for every starting filling and every number of corners, when all corners point outward.



Problem 7: Find a route of flips between these two fillings. Can any two fillings of the same polygon be joined by flips? Explain why your answer works for any polygon with no inward-pointing corners.

